

DECEMBER 29 2022

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J. Acoust. Soc. Am. 152, 3843–3849 (2022)

<https://doi.org/10.1121/10.0014699>



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Measuring tinnitus in pharmaceutical clinical trials^{a)}

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ABSTRACT:

This paper reviews methods and considerations for measuring tinnitus in clinical trials designed to evaluate treatment options using investigational medicinal products. Tests applied in tinnitus-related research and clinical practice have their own measurement purposes, advantages, and limitations. If the characteristics of each test method are well understood, the test can be effectively used in clinical trials. For the accuracy of clinical trial results, it is necessary to use a test tool with verified validity, reliability, and sensitivity. If a test tool that is likely to have high variability in the same individual is required in the clinical trial, strategies to increase the reliability of the test, such as repeat measurements, may also be needed. In addition, a test tool that meets the purpose of the clinical trial should be selected. For example, the tinnitus questionnaire is appropriate to assess reactions to tinnitus, and measurements of tinnitus loudness or pitch are appropriate to evaluate the psychoacoustic characteristics of tinnitus. In conclusion, the use of validated test tools that meet the purpose of the trial will help with the accuracy of the clinical trial results.

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(Received 1 June 2022; revised 17 September 2022; accepted 21 September 2022; published online 29 December 2022)

[Editor: Colleen G. Le Prell]

Pages: 3843–3849

I. INTRODUCTION

Tinnitus is a common symptom worldwide (McCormack *et al.*, 2016). In the United States, the number of individuals with tinnitus is nearly 30×10^6 , accounting for approximately 10% of the total population (Kochkin *et al.*, 2011). The incidence rate among individuals 65–84 years of age is reported to be approximately 26% (Kochkin *et al.*, 2011).

Tinnitus causes a variety of disturbing symptoms and reduces the quality of life. Individuals with tinnitus report various reactions, such as sleep disturbance, and emotional difficulties, such as depression, annoyance, and insecurity (Tyler and Baker, 1983). Some also report difficulty in hearing and concentrating due to tinnitus (Tyler and Baker, 1983; Kochkin *et al.*, 2011).

Many tinnitus sufferers wish to treat their tinnitus, especially with medicinal products (Tyler, 2012). According to a survey of 197 people in a tinnitus self-help group, 52% of the participants responded they would take medicinal products to reduce their symptoms by half, while 30% responded that they would accept “devices (e.g., sound generators).” Further, 62% of the participants responded they would take medicinal products to reduce their symptoms completely, while 42% responded that they would accept “devices” (Tyler, 2012).

At this time, there is no clear cure for tinnitus (Tunkel *et al.*, 2014; McFerran *et al.*, 2019). Various pharmacologic interventions have been developed to treat tinnitus, but a medicinal product that cures tinnitus has not yet been

developed (Cederroth *et al.*, 2018). One reason that medicinal products have not been recommended to treat tinnitus is that there are several subtypes depending on causes and mechanisms, and each subtype requires a different treatment (Stouffer and Tyler, 1990; Tyler *et al.*, 2008; Mohan *et al.*, 2022). It is unlikely that one medicinal product will help everyone. Another limitation of the development of new medicinal products for the treatment of tinnitus is that it is difficult to accurately determine the effect of a new medicinal product due to the lack of objective outcome measurements (Jackson *et al.*, 2019). Currently, most tests that measure the effects of tinnitus are subjective tests (e.g., questionnaires, pitch, and loudness) based on the reaction of the tinnitus sufferer (Tyler *et al.*, 2006). If the effectiveness of a new investigational medicinal product is demonstrated using a subjective measurement tool or method whose reliability and validity have not been verified in clinical trials, the measurement of effectiveness will be unreliable with a limitation in determining the accurate effect.

Although a tool to objectively measure tinnitus has not been developed, various methods for measuring the presence and severity of tinnitus, or to help distinguish subtypes of tinnitus, have been proposed and developed (e.g., Tyler *et al.*, 2006; Tyler *et al.*, 2014; Henry, 2016). In this article, we wish to propose methods and considerations for measuring tinnitus in patients enrolled in clinical trials evaluating the efficacy of investigational medicinal products to cure tinnitus.

II. SCALING

Scaling methods for magnitude estimation have been widely used to measure the degree of human sensation,

^{a)}This paper is part of a special issue on Noise-Induced Hearing Disorders: Clinical and Investigational Tools.

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including the subjective magnitude of an auditory sensation (Stevens, 1946). There are various classifications of scales (e.g., nominal, ordinal, interval, and ratio), each with its own advantages and limitations, but numerical scales are known to be suitable for estimating equal differences and equal ratios or for statistical analysis of auditory sensation (Stevens, 1946).

Among the numerical scaling methods for measuring psychometric responses in humans, the Visual Analog Scale (VAS) is commonly applied (Yeung and Wong, 2019). The VAS, which was introduced as the “graphic rating method” by Hayes and Patterson (1921), is a rating scale that was initially used to gauge participants’ psychometric responses like pain (Yeung and Wong, 2019). VAS has been assessed by many researchers in different fields to evaluate appetite, asthma, craving, dyspepsia, mood, and pain (Wewers and Lowe, 1990; Jamison *et al.*, 2002; Delgado *et al.*, 2018).

Labelling a VAS has important consequences that are not always appreciated. When scores are averaged, within or across individuals, it is important to use a linear scale. However, labelling the VAS can create complications. Typically, labels can be applied to the end points, but labels throughout the scale should not be used. The magnitude between “YES” and “SOMETIMES” might not be the same as the magnitude between “SOMETIMES” and “NO” [as is used in the tinnitus handicap inventory (THI) (Newman *et al.*, 1996)]. Labelling is appropriate at the end points, but not throughout the scale.

If it is necessary to measure the effects of tinnitus of study participants in a clinical trial, the 100-point VAS for loudness and VAS for annoyance can be used with only the end points labeled. Because VAS measures an individual’s subjective psychological state, it is important to use a scale with proven reliability in clinical trials (Tyler *et al.*, 2006). Fortunately, these two VAS scales have been frequently applied in tinnitus studies and their reliability and validity has been verified (Adamchic *et al.*, 2012; Dode *et al.*, 2021; Jin *et al.*, 2021). The estimated minimal clinically important difference (MCID) for the two VAS scales is between 10 and 15 points (Adamchic *et al.*, 2012). The MCID means the smallest improvement in a score, and this can be used for interpretation of the VAS scores in clinical trials (Jaeschke *et al.*, 1989). In addition, although various scaling methods exist, including ordinal scales (e.g., no pain, little pain, lots of pain), a 100-point scale seems reasonable because the decimal system is familiar to most patients and allows for greater sensitivity compared to scales with fewer ranges (e.g., 5- or 7-point scales) (Tyler *et al.*, 2006). The 0–100-point scale is a number range that is often used in daily life, such as when counting numbers or money.

An example of a VAS for loudness (annoyance) is shown in Fig. 1. Using the VAS for loudness as an example, the VAS consists of a 10 cm–long horizontal line, where the left side corresponds to not audible (0 score) and the right side corresponds to extremely loud (100 score). It is necessary to keep the distance between each score constant along the length of the horizontal line. Participants write their own

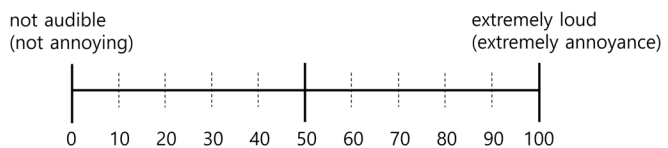


FIG. 1. An example of a VAS for loudness and annoyance.

score or mark a spot on the line that indicates their current level of tinnitus loudness. The VAS can be interpreted that the higher the score, the greater the severity of the corresponding scale. Also, when measuring the change before and after the intervention, a decrease in 15 points or more can be interpreted as a significant improvement based on the estimated MCID (Adamchic *et al.*, 2012). If a validated, reliable, and sensitive scaling method is used, significant changes measured using a VAS in a clinical trial are useful to determine the intervention effect.

III. QUESTIONNAIRES

According to the psychological model of tinnitus, the overall impact of tinnitus is influenced by the characteristics of the tinnitus (e.g., pitch and loudness) and the psychological makeup of each tinnitus sufferer (Dauman and Tyler, 1992; Tyler *et al.*, 1992). Therefore, when evaluating the efficacy of a new investigational medicinal product, it is necessary to distinguish these two aspects and to use an appropriate test tool to evaluate the effect of each aspect (characteristics of the tinnitus or psychological reactions). A tinnitus questionnaire is an appropriate tool for evaluating an individual’s reaction to tinnitus. There are several available questionnaires that have been tested for reliability and validity, such as the Tinnitus Handicap Questionnaire (THQ) (Kuk *et al.*, 1990), Tinnitus Questionnaire (TQ) (Hiller and Goebel, 2004), Tinnitus Functional Index (TFI) (Meikle *et al.*, 2008), and Tinnitus Primary Function Questionnaire (TPFQ) (Tyler *et al.*, 2014).

In addition to reliability and validity, sensitivity is also a key factor in selecting a proper questionnaire for use in a clinical trial (Tyler *et al.*, 2006). When selecting a tinnitus questionnaire, it may be important to choose the questionnaire that includes subcategories that can reflect the purpose for which a clinical trial is intended to measure. For example, the TPFQ focuses on primary disorders that may have secondary effects on work, social components, and quality of life, and consists of subcategories of concentration, hearing, thoughts and emotion, and sleep (Tyler *et al.*, 2014). The TPFQ consists of 20 items with 4 categories (5 items in each category). Each item is rated on a scale from 0 (strongly disagree) to 100 (strongly agree). The total score of each subcategory is calculated by averaging the scores of all sub-items, and the total TPFQ score is calculated by averaging the scores of all items. The TPFQ is interpreted as a significant change when the score changes by 13 points or more (Tyler *et al.*, 2014). Therefore, the TPFQ is an appropriate questionnaire for measuring the intervention effect on

concentration, hearing, thoughts and emotion, and sleep of participants related to tinnitus reactions.

We would like to share an example of how the results of TPFQ can be interpreted in clinical trials (Table I). At pre-treatment, the total TPFQ score, which means overall psychological reactions to tinnitus, is 43.75, and the tinnitus sufferer complained most of difficulty in sleeping (score of 70 points). An example of the result of a re-measurement of TPFQ after intervention for sleep improvement is shown in Table I (see scores at post-treatment). If only the overall score is identified, the improvement of the score is 7.5, and this can be inaccurately interpreted that the intervention had no significant effect. However, the sub-category most suitable for the purpose of the clinical trial is the sleep area. If we focus on the change in the score of the sleep category, there is a significant improvement of 30 points achieved from 70 points at pre-treatment to 40 points post-treatment. When interpreting TPFQ results, it is important to analyze the results for each subcategory separately in addition to the overall score. In other words, it is important to consider questionnaires with subcategories that are relevant to the purpose of the clinical trial. Different questionnaires have different sensitivities. Therefore, checking the sensitivities of each questionnaire and the sub-areas for each clinical trial is required. Again, it is very important to use a questionnaire with a sensitivity that is consistent with the measurement objective of a clinical trial to ensure accurate and reliable research results.

IV. PITCH

Tinnitus pitch can be highly variable within the same individual, and perceived tinnitus is more likely to be a broad spectrum rather than a single tone. It is also more likely to match the frequency domain in which there is generally hearing loss (Henry and Meikle, 2000; Norena *et al.*, 2002). A pitch test can be helpful to classify subtypes according to the individual's tinnitus frequency.

The most prominent pitch test attempts to identify a pitch similar to the patient's prominent sound or the central frequency of the perceived sound spectrum. Although several test methods are available, the two-alternative forced choice (2AFC) method is reliable and can simply be performed with audiometers (Tyler and Conrad-Armes, 1983; Penner and Bilger, 1992; Neff *et al.*, 2019). The pitch test can be conducted as follows. The test is performed in the ipsilateral ear. Two pulsed tones with different octave frequencies are presented alternately and the participant selects the tone closer to their tinnitus. Based on the response, two

additional pulsed tones are presented subsequently, narrowing the frequency range of the two tones. This is repeated until the desired accuracy is obtained.

The pitch test is performed in the ipsilateral ear to avoid bilateral diplacusis (Tyler and Conrad-Armes, 1983). Performing a pitch test in the ipsilateral ear is likely to yield more reliable results compared to measuring in the contralateral ear. As stated previously, measured tinnitus pitch results can vary greatly in the same individual. Therefore, to increase the reliability of the test, it is recommended to repeat the test seven to nine times and use the average value as the participant's pitch test result (Tyler and Conrad-Armes, 1983).

V. LOUDNESS

The loudness of tinnitus is one factor contributing to the annoyance. Although not always the case, the louder the tinnitus is for an individual, the more annoying the tinnitus will be (Tyler *et al.*, 2007). A loudness test can provide some indication of how loud the patient perceives their tinnitus (Fowler, 1938). In a clinical trial, the tinnitus loudness test can be used to measure a participant's psychoacoustic tinnitus loudness change before and after intervention.

A loudness test is often measured at the participant's tinnitus pitch. However, most tinnitus sufferers will have sensorineural hearing loss. Therefore, when performing the tinnitus loudness test, a small increment in intensity could be a large increment in loudness due to loudness recruitment. If the participant is uncomfortable due to the loudness recruitment phenomenon, or if there is concern regarding an error in the test result, the researcher can measure the tinnitus loudness at a frequency other than the tinnitus pitch (Goodwin and Johnson, 1980).

Loudness tests in clinical trials can be performed in various ways (Henry and Meikle, 2000; Henry, 2016), and an example of one applicable method is as follows (Tyler *et al.*, 2007). Using a pulsed pure tone with the frequency of minimal hearing loss (if available), the stimulus is presented to the ear with no tinnitus (or the ear where the tinnitus is least dominant). The participant is informed that pulsed tones will be presented starting at a low level. The participant should be able to rate the loudness of each tone on a scale from 0 to 100, where 0 represents the softest tone they can imagine and 100 represents the loudest tone they can imagine. The test is started at a low loudness level (10 dB sensation level for normal thresholds). The intensity is increased by 2 dB and presented again. The participant is then asked to rate the loudness on the 0–100 scale. The test is continued

TABLE I. An example of TPFQ score results at pre- and post-treatment.

	TPFQ				
	Concentration	Hearing	Thought and Emotion	Sleep	Total
Pre-treatment	40	35	30	70	43.75
Post-treatment	45	35	25	40	36.25

until the participant finds a loudness “equal” to the loudness of their tinnitus.

It should be noted that psychoacoustic measurements (e.g., loudness or pitch) do not evaluate tinnitus reactions (Dauman and Tyler, 1992; Tyler *et al.*, 1992; Manning *et al.*, 2019). For measuring tinnitus reactions, valid questionnaires like the TPFQ are more appropriate.

VI. MASKING

The minimum masking level (MML) has been used as an indicator in clinical trials of tinnitus for many years (Vernon, 1977; Tyler and Conrad-Armes, 1984). The MML refers to the minimum level of broadband noise (BBN) or octave noise required to make a tinnitus sufferer’s tinnitus inaudible (Mancini *et al.*, 2020). Measurements of the MML are often used to determine the usefulness of a tinnitus masker or level of sound therapy (Jastreboff *et al.*, 1994; Tyler *et al.*, 2012) but can be used as a useful test tool to classify subtypes of tinnitus based on the masking characteristics of participants in a clinical trial.

The MML test can be conducted as follows (Tyler *et al.*, 1984; Mancini *et al.*, 2020). It uses octave bandwidth noise, or BBN, to present the stimuli from low levels (i.e., 20 dB sound pressure level) to the ear with greater tinnitus. Using an ascending method, the intensity is increased in 2 dB increments to find a level at which the patient believes their tinnitus is masked. The test can be repeated twice or thrice, and if the test results are within 5 dB, the average value of the MML tests can be considered the final MML result.

There are several considerations for using the MML test in clinical trials. The effects of MML can be very diverse: some participants’ tinnitus is easily masked but some are difficult to mask, or some can be perceived as too loud although they are masked (Tyler *et al.*, 2012; Henry, 2016). Also, there does not appear to be a clear correlation between MML and tinnitus reactions (Tyler *et al.*, 2006; Henry, 2016). For example, even if a tinnitus sufferer has a high MML, difficulty with tinnitus may not be severe. Therefore, the MML and whether it can be masked may be important for subtypes.

VII. POST MASKING (i.e., RESIDUAL INHIBITION)

An individual’s tinnitus may recede for some time after the masker is turned off, which is referred to as post masking (Vernon and Schleuning, 1978). After the masker is turned off, tinnitus may return to normal immediately, may be absent for a period of time and then gradually return to normal, may only be absent briefly before suddenly returning to normal, or may be louder than normal before gradually returning to normal (Tyler *et al.*, 1984). These post-masking metrics can be useful for assigning subgroups based on post masking effectiveness in the clinical trial.

Figure 2 shows the post masking measurement method and an example of tinnitus returning after post masking. BBN is presented bilaterally to the individual’s ear at an

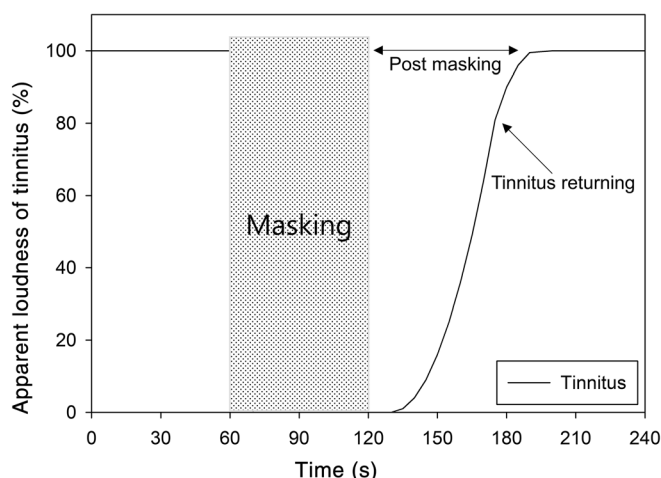


FIG. 2. An example of post masking referenced in Vernon and Schleuning (1978).

intensity of 10 dB higher than the MML for approximately 60 s (Tyler *et al.*, 1984). If the individual experiences suppression of tinnitus after the presentation of BBN, the time from when the tinnitus is suppressed to when the tinnitus returns to normal loudness is measured. Most individuals will experience their tinnitus returning to normal after a brief period after masking; however, there may be a very long duration during when there is a decrease in the loudness of the tinnitus in some cases (Tyler *et al.*, 1984). In some individuals, the tinnitus may become louder (Tyler *et al.*, 1984). Therefore, the stimulus must not be presented for a prolonged duration when measuring post-masking.

VIII. OBJECTIVE MEASUREMENT OF TINNITUS

Many studies have been conducted to develop methods for objectively measuring the presence and degree of tinnitus using imaging and electrophysiology measures, including otoacoustic emissions, electroencephalography, and functional magnetic resonance imaging (fMRI) (e.g., Norton *et al.*, 1990; Weiler and Brill, 2004; Gopal *et al.*, 2017). However, a definitive method for objectively measuring tinnitus has not yet been developed due to the lack of randomized controlled trials, small sample size, and discrepancies between outcome data from different studies (Tunkel *et al.*, 2014; Jackson *et al.*, 2019). If these limitations are addressed, the aforementioned objective measures can be potential surrogate end points for the measurement of tinnitus. A surrogate end point is defined as “an end point that is used in clinical trials as a substitute for a direct measure of how a patient feels, functions, or survives. A surrogate end point does not measure the clinical benefit of primary interest in and of itself, but rather is expected to predict clinical benefit or harm based on epidemiologic, therapeutic, pathophysiology, or other scientific evidence (Food and Drug Administration-National Institutes of Health Biomarker Working Group, 2021).” The U.S. Food and Drug Administration (FDA) currently considers imaging measures to be surrogate end points in some areas like cancer

therapeutics (U.S. FDA, 2018; Chen *et al.*, 2020). Therefore, if a reliable objective tinnitus measure is developed through additional research, it can be used as a surrogate end point in clinical trials.

IX. CLINICALLY MEANINGFUL BENEFITS IN MEASURING PATIENT-REPORTED OUTCOMES

The statistical significance of a treatment effect is influenced by the sample size, and it does not necessarily translate to a clinically significant effect (de Vet *et al.*, 2006). In addition, the U.S. FDA emphasizes “clinically meaningful treatment benefits” in the selection of patient-reported outcome measures (U.S. Food and Drug Administration, 2009). Therefore, when selecting patient-reported outcome measures in pharmaceutical clinical trials for tinnitus treatment, it is important to identify the criteria for clinically significant differences.

Several of the patient-reported outcome measures mentioned in this review have suggested criteria for minimum clinically significant differences (Table II). Regarding VAS, a minimum significant improvement is suggested by an improvement of 15 points or more (Adamchic *et al.*, 2012). However, the measurement can be influenced by the length of the line, as well as any markings and labels on the line. Therefore, it is important to comply with the specifications of the VAS where minimum clinically significant differences have been identified. In addition, for tinnitus questionnaires, minimum significant improvement is proposed by a change of 13 points in TPFQ, 21 points in THQ, and 14 points in TFI (Newman *et al.*, 1995; Tyler *et al.*, 2014; Fackrell *et al.*, 2016). Of note, these current proposed criteria may change in the future depending on the results of further studies, such as measures of reliability for minimum clinically significant differences. Therefore, selection of outcome measures in clinical trials for new investigational medicinal products for tinnitus treatment should be guided by the latest research results on clinically significant benefits.

X. THE MAJOR OUTCOME MEASUREMENTS BEING USED IN CLINICAL TRIALS ASSESSING PHARMACEUTICAL INTERVENTIONS FOR TINNITUS

The major outcome measurements being used in clinical trials on pharmaceutical interventions for tinnitus are

TABLE II. Information of proposed clinically significant differences for patient-reported outcome measurements for tinnitus.

Outcome measurements (total score range: 0–100 points)	Proposed clinically significant difference: change of score	References
VAS for loudness or annoyance	Over 15 points	Adamchic <i>et al.</i> , 2012
TPFQ	Over 13 points	Tyler <i>et al.</i> , 2014
THQ	Over 21 points	Newman <i>et al.</i> , 1995
TFI	Over 14 points	Fackrell <i>et al.</i> , 2016

TFI, Tinnitus Functional Index; THQ, Tinnitus Handicap Questionnaire; TPFQ, Tinnitus Primary Function Questionnaire; VAS, Visual Analog Scale.

presented in Table III (clinicaltrials.gov, 2022). A total of 66 studies that have been completed or are ongoing were searched, and the outcome measurements used in two or more studies are listed in Table III. Scaling methods are the most frequently used tools for measuring outcomes. Many studies have selected pure-tone audiometry, middle ear examination, physical examination, and speech perception test to check the participants' ear and hearing condition, although these tools do not evaluate the tinnitus itself. In addition, many studies have chosen participant self-report to measure the outcome. Moreover, questionnaires have been used in several studies to evaluate the quality-of-life factors, such as sleep and emotion, as an outcome measure. In summary, while some studies have used objective tools, such as fMRI, most studies on pharmaceutical interventions for tinnitus use subjective measurements.

Notably, Tyler *et al.* (2020) recently decried that most “quality of life” scales do not include the importance of hearing and communication and other areas affected by tinnitus. In pharmaceutical clinical trials, the impact of the intervention on the quality of life is important, and selecting a validated questionnaire that can measure various quality of life factors (e.g., hearing and communication) can be useful for confirming effective clinical trial results. For instance, the TPFQ can be a good measurement tool for

TABLE III. The major outcome measurements being used in clinical trials assessing pharmaceutical interventions for tinnitus (clinicaltrials.gov).

Number	Outcome measurements	Number of studies (total number of studies: 66)
1	Scaling (e.g., satisfaction, severity, loudness, distress, annoyance, and awareness)	37
2	Tinnitus Handicap Inventory (THI)	27
3	Puretone audiometry	21
4	Patient self-report (e.g., severity, seriousness, and loudness)	19
5	Tinnitus Functional Index (TFI)	13
6	Loudness measurement	10
7	Beck Depression Inventory (BDI)	6
8	Pittsburg Sleep Quality Index (PSQI)	6
9	Pitch measurement	5
10	Tinnitus Handicap Questionnaire (THQ)	5
11	Minimum Masking Level (MML)	4
12	Otoacoustic emission	4
13	World Health Organization Quality of Life (WHOQOL) questionnaire	4
14	Otoscopic examination	3
15	Beck Anxiety Inventory (BAI)	2
16	Functional magnetic resonance imaging (fMRI)	2
17	Tympanometry	2
18	Tinnitus Questionnaire (TQ)	2
19	Words in Noise test	2

various quality of life factors (concentration, hearing, thoughts and emotion, and sleep) and can provide important insight into the clinically meaningful treatment benefits of an investigational medicinal product (13 or more points) (Tyler *et al.*, 2014).

XI. CONCLUSION

This review offers the following suggestions:

- When choosing a test to measure the effects of tinnitus in clinical trials, researchers should consider test tools that are validated, reliable, and sensitive. Outcome measures should be reliable across multiple settings and participants.
- Strategies to increase the reliability of the test, such as repeat measurements, should also be used.
- In clinical trials, a test tool that meets the measurement purpose should be selected.

In this review, we have tried to cover the features, procedures, and precautions of various measurements (scaling, questionnaires, loudness, pitch, masking, and post masking) to measure the effects of tinnitus. The use of validated test tools that meet the purpose of the trial will lead to accurate clinical trial results.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF), funded by the Korea government (MSIT) (Grant No. 2022R1H1A2091291).

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